SMOOTH EMBEDDINGS OF CLOSED SIMPLY CONNECTED 15-MANIFOLDS IN $\mathbb{R}^{27}$

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Abstract. We prove that every closed, simply connected, smooth 15-manifold M embeds smoothly into $\mathbb{R}^{27}$ regardless of whether $w_2(M) = 0$ or not. Since $2 \cdot 27 \geq 3 \cdot 16$, Haefliger's metastable-range theorem applies, reducing the question to constructing a nowhere-zero section of 27λ over the unordered configuration space of M. Since this space has dimension $30 < 2(27-1)$, the section problem lies in the stable range, where the only obstruction to such a section is a class $o \in \pi_3(R)$, with R the relative Thom spectrum of a virtual bundle over the Borel construction on $M \times M$ relative to its diagonal. This class o coincides with the image of the double point class of any self-transverse immersion $M \looparrowright \mathbb{R}^{27}$. By employing a stable splitting, we compute the homology of the pair up to degree 3. For the immersion (f, ℓ), formed by Cohen's immersion $f\colon M \looparrowright \mathbb{R}^{26}$ and a generic function ℓ, we show that the Hurewicz image of o vanishes, which restricts o to $\eta \cdot \pi_2(R)$. Finally, test calculations over S^2 demonstrate that $\eta \cdot \pi_2(R)$ and the image of double point classes intersect only in 0, thereby establishing $o = 0$.

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The main conclusions and proof process of this paper were obtained through exploration by GPT 5.6sol and underwent two rounds of independent review by Opus 5.5. However, since they have not been reviewed by peer experts, accuracy is still not guaranteed. The purpose of publishing this paper is to share and exchange knowledge and to reduce the burden on mathematicians working on an extremely difficult conjecture. Anyone interested is welcome to point out issues in this paper and communicate with me by email: `EvaristoHuang@outlook.com`. Academic credit for the final results should go to those who complete the manual verification and pass the review process.

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2. Introduction

Theorem 2.1. *Every closed, simply connected, smooth 15-manifold M admits a smooth embedding*

$$M^{15} \hookrightarrow \mathbb{R}^{27}.$$

This holds both when $w_2(M) = 0$ and when $w_2(M) \neq 0$.

The only input from immersion theory is the immersion theorem of Cohen [Coh85]. It states that every closed smooth n-manifold immerses in $\mathbb{R}^{2n-\alpha(n)}$, where $\alpha(n)$ is the number of ones in the binary expansion of n. Since $15 = 1111_2$, we have $\alpha(15) = 4$, so every closed M^{15} immerses in $\mathbb{R}^{26}$.

Since $2 \cdot 27 \geq 3(15+1)$, Haefliger's theorem [Hae62] reduces the existence of an embedding to the existence of a nowhere-zero section of 27λ over the unordered configuration space of M. In the stable range, the obstruction to such a section is a class

$$o \in \pi_3(R),$$

where R is a relative Thom spectrum over the pair (B, A) formed by the Borel construction on $M \times M$ and its diagonal. For every self-transverse immersion $g\colon M \looparrowright \mathbb{R}^{27}$, the class $o = \varrho_*\mu(g)$ is the image of the double point class $\mu(g)$ (Section 3). Section 4 computes the homology of (B, A) through degree 3 by means of a stable splitting. Section 5 shows that the Hurewicz image of o vanishes for $g = (f, \ell)$, where f is a Cohen immersion and ℓ is a generic function. It follows that $o \in \eta \cdot \pi_2(R)$. Section 6 then shows, by test computations over S^2, that $\eta \cdot \pi_2(R)$ meets the image of ϱ_* only in 0. The remark at the end of Section 6 assembles the proof.

Unless stated otherwise, M denotes a closed, connected, simply connected smooth manifold of dimension 15 with basepoint x_0, and we write

$$v := w_2(M) \in H^2(M; \mathbb{F}_2).$$

Because M is simply connected, it is orientable, $H_1(M; \mathbb{Z}) = 0$ and $H_2(M; \mathbb{F}_2) = H_2(M; \mathbb{Z}) \otimes \mathbb{F}_2$. The group $C_2 = \{1, \tau\}$ acts on $M \times M$ by $\tau(x, y) = (y, x)$. Let $EC_2 \to BC_2 = \mathbb{RP}^\infty$ be the universal double cover, let λ be the associated real line bundle on any Borel construction, and let $u = w_1(\lambda)$.

We work in the stable homotopy category. Here $\mathbb{S}$ is the sphere spectrum, π_* denotes stable homotopy groups, and $\Sigma^\infty K_+$ is the suspension spectrum of a space K. For a pointed space K, $\Sigma^\infty K$ denotes its reduced suspension spectrum, and S^k also denotes the sphere spectrum $\Sigma^k\mathbb{S}$. All homology and cohomology groups of spectra are reduced.

Fact 2.2 (Stable stems, [Tod62])**.** *$\pi_0(\mathbb{S}) = \mathbb{Z}$, $\pi_1(\mathbb{S}) = \mathbb{Z}/2\{\eta\}$ and $\pi_2(\mathbb{S}) = \mathbb{Z}/2\{\eta^2\}$.*

Fact 2.3 (Thom spectra)**.** *For a virtual vector bundle ζ of rank k over a CW complex K with a subcomplex L, we write K^ζ for the Thom spectrum and set $(K, L)^\zeta = \operatorname{cofib}(L^{\zeta|L} \to K^\zeta)$. These constructions are natural for maps of pairs covered by bundle maps. If K has a single 0-cell, then K^ζ has a CW structure with one $(k+i)$-cell for each i-cell of K. Its bottom cell is the inclusion ι_0 of the Thom spectrum of a fibre. There is a mod 2 Thom isomorphism $H^{*+k}\big((K, L)^\zeta; \mathbb{F}_2\big) = U \cdot H^*(K, L; \mathbb{F}_2)$. If $w_1(\zeta) = 0$, a choice of orientation also gives an integral Thom isomorphism*

$$H_{*+k}\big((K, L)^\zeta; \mathbb{Z}\big) \cong H_*(K, L; \mathbb{Z}).$$

These isomorphisms are compatible with mod 2 reduction, with connecting homomorphisms, and with maps of pairs covered by bundle maps. Let Z be a closed manifold of dimension $-k$, with a map $Z \to K$ and a stable isomorphism between its stable normal bundle and the pullback of ζ. Under the Thom isomorphism, the Hurewicz image of its Pontryagin–Thom class

in $H_0(K^\zeta)$ is then the image of the fundamental class of Z in $H_{-k}(K)$. With $\mathbb{Z}$ coefficients, Z is oriented by the orientation of ζ; with $\mathbb{F}_2$ coefficients, no orientation is needed. Since $\mathrm{Sq}^i U = w_i(\zeta)U$ [MS74, §8]*, the Cartan formula gives, for $a \in H^*(K, L; \mathbb{F}_2)$,*

$$\mathrm{Sq}^2(U \cdot a) = U \cdot \big(\mathrm{Sq}^2 a + w_1(\zeta)\,\mathrm{Sq}^1 a + w_2(\zeta)\, a\big). \tag{2.4}$$

Lemma 2.5. *Let $P = \mathbb{S} \cup_\psi e^2$ with $\psi \in \pi_1(\mathbb{S})$.*

(a) $\mathrm{Sq}^2\colon H^0(P;\mathbb{F}_2) \to H^2(P;\mathbb{F}_2)$ *is nonzero if and only if $\psi = \eta$.*
(b) *If ζ is a rank-0 virtual bundle over S^2 with $w_2(\zeta) \neq 0$, then $(S^2)^\zeta \simeq \mathbb{S} \cup_\eta e^2$ and $\eta\,\iota_0 = 0$ in $\pi_1\big((S^2)^\zeta\big)$.*

Proof. (a) If $\psi = 0$, then $P \simeq \mathbb{S} \vee S^2$. In this case the generator of H^0 is pulled back along the retraction $P \to \mathbb{S}$, and $H^2(\mathbb{S};\mathbb{F}_2) = 0$, so $\mathrm{Sq}^2 = 0$. If $\psi = \eta$, then $P \simeq \Sigma^{-2}\Sigma^\infty\mathbb{CP}^2$. In this case Sq^2 of the degree-2 generator of $H^*(\mathbb{CP}^2;\mathbb{F}_2)$ is its square, which is nonzero. (b) The CW structure of S^2 with one 0-cell and one 2-cell gives $(S^2)^\zeta = \mathbb{S} \cup_\psi e^2$, with bottom cell ι_0. By (2.4), $\mathrm{Sq}^2 U = w_2(\zeta)U \neq 0$, so $\psi = \eta$ by (a). The attaching map $\eta\,\iota_0$ is therefore null in $(S^2)^\zeta$. □

Lemma 2.6. *Let K be a connected CW complex and ζ a rank-0 virtual bundle over K with $w_1(\zeta) = w_2(\zeta) = 0$. Then $\eta\,\iota_0 \neq 0$ in $\pi_1(K^\zeta)$ and $\eta^2\iota_0 \neq 0$ in $\pi_2(K^\zeta)$.*

Proof. Because $w_1(\zeta) = w_2(\zeta) = 0$, the classifying map $K \to BO$ of ζ lifts to BSpin. Passing to Thom spectra gives a map of spectra $K^\zeta \to \mathrm{MSpin}$ that sends ι_0 to the image of the unit $\mathbb{S} \to \mathrm{MSpin}$. Since BSpin is 3-connected, MSpin has a CW structure with one 0-cell and all other cells of dimension ≥ 4. The unit is therefore an isomorphism on π_1 and π_2. By Fact 2.2, η and η^2 map to nonzero elements of $\pi_*(\mathrm{MSpin})$, so $\eta\,\iota_0$ and $\eta^2\iota_0$ are nonzero. □

Lemma 2.7. *Let $\Sigma^2\mathbb{S}/2 = S^2 \cup_2 e^3$ be the double suspension of the mod 2 Moore spectrum, with bottom cell $i_0\colon S^2 \to \Sigma^2\mathbb{S}/2$. Then $\pi_3(\Sigma^2\mathbb{S}/2) = \mathbb{Z}/2\{i_0\eta\}$.*

Proof. The cofibre sequence $S^2 \xrightarrow{2} S^2 \xrightarrow{i_0} \Sigma^2\mathbb{S}/2 \to S^3 \xrightarrow{2} S^3$ gives the exact sequence $\pi_3(S^2) \xrightarrow{2} \pi_3(S^2) \xrightarrow{(i_0)_*} \pi_3(\Sigma^2\mathbb{S}/2) \to \pi_3(S^3) \xrightarrow{2} \pi_3(S^3)$. The first map is zero on $\pi_3(S^2) = \mathbb{Z}/2\{\eta\}$, and the last map is injective on $\mathbb{Z}$. □

Lemma 2.8 (Cellular Hurewicz)**.** *Let P be a CW spectrum with cells only in dimensions $\geq k$. If $y \in \pi_{k+1}(P)$ has zero Hurewicz image in $H_{k+1}(P;\mathbb{Z})$, then $y \in \eta \cdot \pi_k(P)$.*

Proof. Since $P/P^{(k+1)}$ has cells only in dimensions $\geq k+2$, the map $\pi_{k+1}(P^{(k+1)}) \to \pi_{k+1}(P)$ is surjective, so y lifts to some $\tilde{y}$. The quotient map $\varpi\colon P^{(k+1)} \to P^{(k+1)}/P^{(k)} = \bigvee S^{k+1}$ sends $\tilde{y}$ to its cellular chain $\varpi_*\tilde{y} \in \pi_{k+1}(\bigvee S^{k+1}) = C_{k+1}$. This chain is a cycle representing $h(y)$. As $h(y) = 0$, we have $\varpi_*\tilde{y} = d\beta$ for some $\beta \in C_{k+2}$, where d is the cellular boundary. The corresponding combination $a_\beta \in \pi_{k+1}(P^{(k+1)})$ of attaching maps of $(k+2)$-cells satisfies $\varpi_* a_\beta = d\beta$ and maps to 0 in $\pi_{k+1}(P)$. Thus $\tilde{y} - a_\beta$ lifts y and lies in the kernel of ϖ_*, so it comes from $\pi_{k+1}(P^{(k)})$. Since $P^{(k)} = \bigvee S^k$, every element of $\pi_{k+1}(P^{(k)})$ is a sum of elements $\eta\, e$, where e is the inclusion of a k-cell. Hence its image lies in $\eta \cdot \pi_k(P)$. □

Fact 2.9 (Minimal cell structures)**.** *Let $P = \Sigma^{-m}\Sigma^\infty P'$, where P' is a simply connected finite CW complex, and suppose that $H_j(P;\mathbb{Z}) = 0$ for $j \leq 1$. Then P is equivalent to a CW spectrum having one j-cell for each $\mathbb{Z}$ summand of $H_j(P;\mathbb{Z})$, and a j-cell and a $(j+1)$-cell for each $\mathbb{Z}/k$ summand of $H_j(P;\mathbb{Z})$, in a chosen cyclic decomposition. In particular, P has no cells of dimension < 2. This follows by applying* [Hat02, Prop. 4C.1] *to P'.*

3. Reduction to a double point obstruction

The configuration space. In this section M may be any closed smooth manifold; we apply the results to S^2 in Section 6. The normal bundle of the diagonal $\Delta \subset M \times M$ is TM, and τ acts on it by -1. Fix a Riemannian metric on M. For $\epsilon_0 > 0$ below the injectivity radius, $N(\Delta) = \{(x, y) : d(x, y) \leq \epsilon_0\}$ is a closed τ-invariant tubular neighbourhood of Δ, equivariantly diffeomorphic to $D(TM)$. Put

$$\widetilde{W} = \overline{M \times M \setminus N(\Delta)}, \qquad W = \widetilde{W}/C_2.$$

Then $\widetilde{W}$ is a compact manifold with free C_2-action and $\partial\widetilde{W} = S(TM)$. Hence W is a compact manifold of dimension $2 \dim M$ with $\partial W = S(TM)/C_2$. Radial deformation exhibits $\widetilde{W}$ as a C_2-equivariant deformation retract of $M \times M \setminus \Delta$. Different admissible ϵ_0 give equivariantly isotopic $\widetilde{W}$, compatibly with all constructions below. We shrink ϵ_0 once, in the paragraph on double points.

Theorem 3.1 (Haefliger [Hae62])**.** *Let M^n be a closed smooth manifold and $2N \geq 3(n + 1)$. If there is a C_2-equivariant map $M \times M \setminus \Delta \to S^{N-1}$, where C_2 acts on S^{N-1} by the antipodal map, then M embeds smoothly in $\mathbb{R}^N$.*

For $n = 15$ and $N = 27$ we have $54 \geq 48$. An equivariant map $\widetilde{W} \to S^{26}$ is the same as a nowhere-zero section of 27λ over W. We therefore obtain the following criterion.

Corollary 3.2. *If $\dim M = 15$ and 27λ admits a nowhere-zero section over W, then M embeds smoothly in $\mathbb{R}^{27}$.*

The stable range. For a real vector bundle V over a space X, let V^+ be its fibrewise one-point compactification. It is a bundle of spheres with two sections, namely the zero section and the section at infinity. Write $S^k = \mathbb{R}^k \cup \{\infty\}$ and $\mathrm{pr}\colon X \times \mathbb{R}^k \to X$. A *pointed fibrewise map* $X \times S^k \to (V \oplus \mathbb{R}^k)^+$ is a map over X that sends $X \times \{\infty\}$ to the section at infinity. The stable cohomotopy group $\omega^0(X; -V)$ of [Cra80, §2] is represented by such maps, $k \geq 0$, modulo pointed fibrewise homotopy and fibrewise suspension. Its zero is the class of the section at infinity. The Euler class $\gamma'(V) \in \omega^0(X; -V)$, denoted $\gamma(V)$ in [Cra80, (2.1)], is the class of the pointed fibrewise map $X \times S^0 \to V^+$ given by the zero section.

Let X be a compact smooth manifold and ζ a virtual bundle on X. We write $\Omega_m(X, \partial X; \zeta)$ for the set of bordism classes of compact m-manifolds Z with a map $f\colon (Z, \partial Z) \to (X, \partial X)$ and a stable isomorphism $\nu_Z \cong f^*\zeta$ of the stable normal bundle. We require that $f^{-1}(\partial X) = \partial Z$ and that f is compatible with collars near ∂Z. Bordisms are compact manifolds with corners mapping to $(X \times [0, 1], \partial X \times [0, 1])$ under the same conditions. After a homotopy of f, every class has such a representative.

Lemma 3.3 (Fibrewise Pontryagin–Thom construction)**.** *Let X be a compact smooth d-manifold, possibly with boundary, and $V \to X$ a real vector bundle of rank r.*

(a) *Every element of $\omega^0(X; -V)$ is represented by a pointed fibrewise map $F\colon X \times S^k \to (V \oplus \mathbb{R}^k)^+$ that is smooth and transverse to the zero section near $F^{-1}(0)$, also along $\partial X \times \mathbb{R}^k$. For such F, $Z_F = F^{-1}(0) \subset X \times \mathbb{R}^k$ is a compact neat submanifold of dimension $d - r$. The derivative dF identifies $\nu(Z_F \subset X \times \mathbb{R}^k)$ with $(V \oplus \mathbb{R}^k)|_{Z_F}$, and hence ν_{Z_F} with $(V - TX)|_{Z_F}$. The assignment $F \mapsto [Z_F]$ is a well-defined bijection*

$$\omega^0(X; -V) \longrightarrow \Omega_{d-r}(X, \partial X; V - TX)$$

that sends the zero of $\omega^0(X; -V)$ to the class of the empty manifold.

(b) *The Pontryagin–Thom construction gives a bijection* $\Omega_{d-r}(X, \partial X; V - TX) \cong \pi_0\big((X, \partial X)^{V-TX}\big)$ *that sends the empty manifold to* 0.
(c) *Let* s *be a section of* V *transverse to the zero section, also along* ∂X*. Then the bijections of* (a) *and* (b) *send* $\gamma'(V)$ *to the class* $[Z(s)]$ *of* $Z(s) = s^{-1}(0)$*, with the normal data given by* ds*.*

Proof. (a) *Transversality.* Let F be a pointed fibrewise map. The set $F^{-1}(0)$ of points sent to the zero section is closed in $X \times S^k$ and disjoint from $X \times \{\infty\}$. It is therefore a compact subset of $X \times \mathbb{R}^k$. Choose a compact neighbourhood K of $F^{-1}(0)$ on which F is finite, that is, a section of the vector bundle $\mathrm{pr}^*(V \oplus \mathbb{R}^k)$, and let $\epsilon = \min_{\partial K} |F| > 0$. A homotopy of sections supported in K and of size $< \epsilon$ creates no new zeros near ∂K. Let ρ be a cut-off function supported in K and equal to 1 near $F^{-1}(0)$, and let the sections σ_i span the fibres over K. Smooth approximation and the transversality theorem of [GP74, Ch. 2], applied to the family $F + \rho \sum_i t_i \sigma_i$, give such a homotopy. It makes F smooth and transverse to zero near $F^{-1}(0)$, also along $\partial X \times \mathbb{R}^k$.

The zero set. For transverse F, Z_F is a compact neat submanifold of codimension $r + k$ in $X \times \mathbb{R}^k$, so $\dim Z_F = d - r$. The derivative dF identifies $\nu(Z_F \subset X \times \mathbb{R}^k)$ with $(V \oplus \mathbb{R}^k)|_{Z_F}$. Since $T(X \times \mathbb{R}^k) = TX \oplus \mathbb{R}^k$, this gives $TZ_F \oplus V \oplus \mathbb{R}^k \cong TX \oplus \mathbb{R}^k$ on Z_F, that is, a stable isomorphism $\nu_{Z_F} \cong (V - TX)|_{Z_F}$.

Well defined. Consider a pointed fibrewise homotopy between two transverse maps, made transverse relative to its ends by the same argument. Its zero set is a bordism between their zero sets. Fibrewise suspension replaces Z_F by $Z_F \times \{0\} \subset X \times \mathbb{R}^{k+1}$, which has the same class. The section at infinity has empty zero set.

Collapse maps. Let $Z \subset X \times \mathbb{R}^k$ be a compact neat submanifold with an isomorphism $\iota\colon \nu(Z \subset X \times \mathbb{R}^k) \cong (V \oplus \mathbb{R}^k)|_Z$. Choose a closed tubular neighbourhood $U \cong D(\nu)$ of Z, neat along $\partial X \times \mathbb{R}^k$. Choose also an isomorphism Φ over U from the pullback of $(V \oplus \mathbb{R}^k)|_Z$ along the projection $U \to Z$ to $\mathrm{pr}^*(V \oplus \mathbb{R}^k)|_U$, equal to the identity over Z. Let $u \in \operatorname{int} U$ correspond to $\xi \in \nu_z$ with $|\xi| < 1$. The collapse map c sends u to $\Phi_u\big(\iota(\xi)/(1 - |\xi|)\big) \in (V \oplus \mathbb{R}^k)_{\mathrm{pr}(u)}$ and sends the complement of $\operatorname{int} U$ to infinity. It is a pointed fibrewise map, transverse to zero, with zero set Z and normal data ι (cf. the Pontryagin construction in [Mil65, §7]).

Surjectivity. Let $(Z, f, \nu_Z \cong f^*(V - TX))$ represent a class. For k large there is a map $e\colon Z \to \mathbb{R}^k$ such that $(f, e)\colon Z \to X \times \mathbb{R}^k$ is a neat embedding. Its normal bundle ν satisfies $TZ \oplus \nu = (TX \oplus \mathbb{R}^k)|_Z$. The given stable isomorphism therefore becomes a stable isomorphism $\nu \cong (V \oplus \mathbb{R}^k)|_Z$. Since $O(n) \to O$ is $(n-1)$-connected, for k large it is induced by a genuine isomorphism ι. The collapse map of (Z, ι) represents the given class.

Injectivity. Let F be transverse and $Z = Z_F$, with $\iota = dF$ and with U and Φ as above. On U put $L(\xi) = \Phi\big(\iota(\xi)\big)$. The maps F and L vanish on Z with the same derivative. After rescaling the metric on ν, we may therefore assume that F is finite and $|F - L| \le \frac{1}{2}|L|$ on the disc bundle U_2 of radius 2. Let ρ be a cut-off equal to 1 on U_1 and supported in U_2. Then $F_t = F + t\rho(L - F)$ is a pointed fibrewise homotopy with no zeros off Z, ending at F_1 with $F_1 = L$ on U_1. Let c be the collapse map formed with U_1. On $U_{1/2}$, $H_t(\xi) = \big(1 - t + t/(1 - |\xi|)\big)L(\xi)$ joins F_1 to c without new zeros. Now consider the closed set $C = X \times S^k \setminus \operatorname{int} U_{1/2}$. On it, the maps F_1, c and $H_t|_{\partial C}$ take values in $(V \oplus \mathbb{R}^k)^+$ minus the zero section. Via $v \mapsto v/|v|^2$, this space is a vector bundle over X whose zero section is the section at infinity. Hence the section given on $C \times \{0, 1\} \cup (\partial C \cup X \times \{\infty\}) \times [0, 1]$ extends over $C \times [0, 1]$, and $F \simeq c$.

Suppose now that $[Z_F] = [Z_{F'}]$, and suspend both maps to a common large k. A bordism Z_I between the zero sets embeds neatly in $X \times [0, 1] \times \mathbb{R}^k$ over its map to $X \times [0, 1]$, extending the given embeddings of its ends. Since $r + k > \dim Z_I + 1$, its normal data are induced by a genuine

isomorphism extending dF and dF'. Form the collapse map with a tubular neighbourhood and an isomorphism over it that extend those at the ends. It is a pointed fibrewise homotopy between collapse maps of Z_F and $Z_{F'}$. Hence $F \simeq F'$ stably.

(b) This is the Pontryagin–Thom theorem for normal bordism [Sto68, Ch. II], in relative form. Since X is compact, for N large $V - TX + N$ is represented by a vector bundle ζ_N of rank $N - \dim Z$. Embed $(Z, \partial Z)$ neatly in (D^N, S^{N-1}). For N large, the normal data give an isomorphism of its normal bundle with $f^*\zeta_N$. The collapse $(D^N, S^{N-1}) \to (\operatorname{Th}\zeta_N, \operatorname{Th}\zeta_N|_{\partial X})$ then defines an element of $\pi_N\big(\operatorname{Th}\zeta_N/\operatorname{Th}\zeta_N|_{\partial X}\big) = \pi_0\big((X,\partial X)^{V-TX}\big)$. Transversality to the zero section gives the inverse, as for closed manifolds. The rank of $V - TX$ is $r - d = -\dim Z$, so the class lies in π_0.

(c) The maps $X \times S^0 \to V^+$ given by the sections ts, $0 \le t \le 1$, are pointed fibrewise maps. The zero section is therefore homotopic to s. The section s is transverse with zero set $Z(s)$, so it represents $[Z(s)]$ under (a) with $k = 0$. □

Proposition 3.4. *Let X be a compact smooth d-manifold, possibly with boundary, and $V \to X$ a real vector bundle of rank r with $d < 2(r-1)$. Let s be a section of V transverse to the zero section, also along ∂X, and let*

$$\gamma(V) = [Z(s)] \in \pi_0\big((X,\partial X)^{V-TX}\big)$$

be the Pontryagin–Thom class of its zero set with the normal data given by ds. Then $\gamma(V)$ does not depend on s, and V admits a nowhere-zero section if and only if $\gamma(V) = 0$.

Proof. By Lemma 3.3, $\gamma(V)$ is the image of $\gamma'(V)$ under a bijection that sends the zero of $\omega^0(X; -V)$, the class of the section at infinity, to 0. In particular, $\gamma(V)$ does not depend on s. We apply [Cra80, Prop. (2.4)(i)] to the compact ENR X with the closed sub-ENR $Y = \emptyset$. In the range $d < 2(r-1)$, it shows that V has a nowhere-zero section if and only if $\gamma'(V) = 0$. □

The Borel pair. Let

$$B = EC_2 \times_{C_2} (M \times M), \qquad A = BC_2 \times M, \qquad j\colon A \to B,$$

where j is induced by the diagonal. Let T be the vector bundle on B obtained from $TM \times TM$ by the Borel construction, with C_2 interchanging the factors. Inside B put

$$W' = EC_2 \times_{C_2} \widetilde{W}, \qquad \partial W' = EC_2 \times_{C_2} S(TM), \qquad A' = EC_2 \times_{C_2} N(\Delta).$$

Since C_2 acts freely on $\widetilde{W}$, the projection $p_W\colon (W', \partial W') \to (W, \partial W)$ is a homotopy equivalence of pairs, with $\lambda|_{W'} = p_W^*\lambda$ and $T|_{W'} = p_W^*TW$. The neighbourhood $N(\Delta)$ deformation retracts equivariantly onto Δ, on which C_2 acts trivially. Hence $A \subset A'$ is a homotopy equivalence.

Lemma 3.5. *$B = W' \cup_{\partial W'} A'$, and for every virtual bundle ζ on B the maps*

$$(W', \partial W')^\zeta \longrightarrow (B, A')^\zeta \longleftarrow (B, A)^\zeta$$

are equivalences. If $\zeta|_{W'} = p_W^*\zeta_W$, then $(W', \partial W')^\zeta \simeq (W, \partial W)^{\zeta_W}$. Moreover $B/A \simeq W/\partial W$. The projection $(B, A) \to (\mathrm{SP}^2 M, \Delta M)$, where $\mathrm{SP}^2 M = (M \times M)/C_2$, induces a ring isomorphism $H^*(\mathrm{SP}^2 M, \Delta M; \mathbb{F}_2) \to H^*(B, A; \mathbb{F}_2)$ and isomorphisms in relative homology.

Proof. The equivariant decomposition $M \times M = \widetilde{W} \cup_{S(TM)} N(\Delta)$ is a pushout along a cofibration. The Borel construction preserves it, which gives $B = W' \cup_{\partial W'} A'$. Hence $(W', \partial W')^\zeta \to (B, A')^\zeta$ is an equivalence by excision for Thom spectra. The map $(B, A)^\zeta \to (B, A')^\zeta$ is an equivalence because $A \to A'$ is a homotopy equivalence. The statement about ζ_W follows from the equivalence p_W. Similarly, $\mathrm{SP}^2 M = W \cup_{\partial W} N(\Delta)/C_2$. Here $N(\Delta)/C_2$, the bundle of cones on $\partial W \to M$, deformation retracts onto ΔM, so $\mathrm{SP}^2 M/\Delta M \simeq W/\partial W$. The projection

$B \to \mathrm{SP}^2 M$ restricts to p_W on W' and maps A' to $N(\Delta)/C_2$. It therefore induces an equivalence $B/A \simeq B/A' \to \mathrm{SP}^2 M/(N(\Delta)/C_2) \simeq \mathrm{SP}^2 M/\Delta M$. Being a map of pairs, it respects relative cup products. □

The relative Thom spectrum. From now until the end of this section, $\dim M = 15$. Set

$$\xi = 27\lambda - T \ \ (\text{rank } -3), \qquad z = \xi + 3 \ \ (\text{rank } 0),$$

$$\mathcal{E} = B^z, \qquad A^z = A^{j^* z}, \qquad R = (B, A)^z = \operatorname{cofib}\big(A^z \xrightarrow{j} \mathcal{E}\big),$$

with $\varrho\colon \mathcal{E} \to R$ and connecting map $\delta\colon R \to \Sigma A^z$. They give the exact sequence

$$\pi_k(A^z) \xrightarrow{j_*} \pi_k(\mathcal{E}) \xrightarrow{\varrho_*} \pi_k(R) \xrightarrow{\delta} \pi_{k-1}(A^z) \xrightarrow{j_*} \pi_{k-1}(\mathcal{E}). \tag{3.6}$$

The maps j_*, ϱ_* and δ are induced by maps of spectra and hence commute with multiplication by η. Since $\xi|_{W'} = p_W^*(27\lambda - TW)$, Lemma 3.5 gives

$$\pi_3(R) \cong \pi_3\big((W, \partial W)^{27\lambda - TW + 3}\big) = \pi_0\big((W, \partial W)^{27\lambda - TW}\big). \tag{3.7}$$

Double points. Let $g\colon M \looparrowright \mathbb{R}^{27}$ be a self-transverse immersion. Choose ϵ_0 so small that $g(x) \neq g(y)$ whenever $0 < d(x, y) \leq \epsilon_0$. The map $s_g(x, y) = g(x) - g(y)$ satisfies $s_g \circ \tau = -s_g$, so it is a section of 27λ over W. It has no zeros on $N(\Delta) \setminus \Delta$, and in particular none on ∂W. At (x, y) with $g(x) = g(y)$,

$$ds_g(a, b) = dg_x(a) - dg_y(b), \qquad (a, b) \in T_x M \oplus T_y M.$$

Thus s_g is transverse to the zero section if and only if $dg(T_x M) + dg(T_y M) = \mathbb{R}^{27}$ at every double point. This holds because g is self-transverse. Hence

$$\widetilde{D}_g = \{(x, y) : x \neq y, \ g(x) = g(y)\}, \qquad D_g = \widetilde{D}_g/C_2 = s_g^{-1}(0) \subset \operatorname{int} W$$

are closed 3-manifolds. The derivative ds_g identifies $\nu(D_g \subset W)$ with $27\lambda|_{D_g}$, and hence ν_{D_g} with $(27\lambda - TW)|_{D_g}$.

Definition 3.8. Choose an equivariant map $\widetilde{D}_g \to EC_2$. Together with $\widetilde{D}_g \subset \widetilde{W}$, it defines $\kappa_D\colon D_g \to W' \subset B$ with $p_W \kappa_D$ the inclusion. The *double point class* $\mu(g) \in \pi_3(\mathcal{E}) = \pi_0(B^\xi)$ is the Pontryagin–Thom class of κ_D with the normal data above, and

$$o(g) = \varrho_* \mu(g) \in \pi_3(R).$$

Lemma 3.9. *Under* (3.7)*, $o(g) = \gamma(27\lambda|_W)$.*

Proof. Apply Proposition 3.4 to $X = W$, $V = 27\lambda$ and the section s_g, which is transverse to zero and has no zeros near ∂W. Its zero set D_g is closed and lies in the interior. Hence $\gamma(27\lambda|_W)$ is the image of $[D_g] \in \pi_0(W^{27\lambda - TW})$ in the relative group. By naturality of $W \xleftarrow{p_W} W' \subset B \to (B, A') \leftarrow (B, A)$ and Lemma 3.5, this image is $\varrho_* \mu(g)$. □

Corollary 3.10. *If $o(g) = 0$ for one self-transverse immersion $g\colon M^{15} \looparrowright \mathbb{R}^{27}$, then M embeds smoothly in $\mathbb{R}^{27}$.*

Proof. By Lemma 3.9, $\gamma(27\lambda|_W) = 0$. Proposition 3.4 applies with $d = 30$, $r = 27$ and $30 < 52$, so 27λ has a nowhere-zero section over W. Corollary 3.2 then gives the embedding. □

4. The Borel pair in low degrees

In Lemma 4.1 and Proposition 4.2, M may be any closed, connected, simply connected manifold with basepoint x_0; we apply these results to S^2 in Section 6. From Lemma 4.3 onwards, $\dim M = 15$.

A stable splitting. Write $\mathcal{Y} = \Sigma^\infty M_+$ and $\mathcal{X} = \Sigma^\infty M$, reduced at x_0. Let $c\colon \mathcal{Y} \to \mathbb{S}$ be induced by $M \to \mathrm{pt}$, and let $q\colon \mathcal{Y} \to \mathcal{X}$ be induced by $M_+ \to M$, which sends $+$ to x_0. Let $i\colon \mathbb{S} \to \mathcal{Y}$ be induced by x_0, and let $\mathfrak{s}\colon \mathcal{X} \to \mathcal{Y}$ be a stable section with $q\mathfrak{s} = 1$ and $c\mathfrak{s} = 0$. Then $(c, q)\colon \mathcal{Y} \simeq \mathbb{S} \vee \mathcal{X}$. For a spectrum P put $D_2P = (P \wedge P)_{hC_2} = (EC_{2+} \wedge P \wedge P)/C_2$, with C_2 interchanging the factors. Then

$$\Sigma^\infty B_+ = D_2\mathcal{Y}, \qquad \Sigma^\infty A_+ = \Sigma^\infty BC_{2+} \wedge \mathcal{Y}.$$

The map $\Sigma^\infty j_+$ is induced on homotopy orbits by the diagonal $\Delta\colon \mathcal{Y} \to \mathcal{Y} \wedge \mathcal{Y}$, which is equivariant for the trivial action on $\mathcal{Y}$.

For a spectrum P with trivial action, let $\nabla_P\colon P \to C_{2+} \wedge P = P \vee P \simeq P \times P$ be the diagonal. It is C_2-equivariant for the involution exchanging the two copies. It corresponds to 1_P under the adjunction between restriction and coinduction, $C_{2+} \wedge P \simeq F(C_{2+}, P)$. Define $\mathrm{tr}\colon \Sigma^\infty BC_{2+} \to \mathbb{S}$ as the map induced by $\nabla_{\mathbb{S}}$ on homotopy orbits, followed by $(C_{2+} \wedge \mathbb{S})_{hC_2} = \Sigma^\infty EC_{2+} \simeq \mathbb{S}$. Then $\mathrm{tr} \wedge 1_P$ is the map induced by ∇_P on homotopy orbits. Let $i_1\colon M \to B$ be given by $i_1(x) = [e_0, (x, x_0)]$.

Lemma 4.1. *There are equivalences*

$$\Sigma^\infty B_+ \simeq \Sigma^\infty BC_{2+} \vee \mathcal{X} \vee D_2\mathcal{X}, \qquad \Sigma^\infty A_+ \simeq \Sigma^\infty BC_{2+} \vee \big(\Sigma^\infty BC_{2+} \wedge \mathcal{X}\big),$$

in which the first summand of $\Sigma^\infty B_+$ is projected to by $B \to BC_2$, the composite $(\Sigma^\infty i_{1+}) \circ \mathfrak{s}\colon \mathcal{X} \to \Sigma^\infty B_+$ is the inclusion of the summand $\mathcal{X}$, and

$$\Sigma^\infty j_+ = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{tr} \wedge 1_{\mathcal{X}} \\ 0 & \chi \end{pmatrix}$$

for some map $\chi\colon \Sigma^\infty BC_{2+} \wedge \mathcal{X} \to D_2\mathcal{X}$.

Proof. The smash square of (c, q) is an equivariant equivalence

$$\mathcal{Y} \wedge \mathcal{Y} \simeq \mathbb{S} \ \vee\ \big((\mathbb{S} \wedge \mathcal{X}) \vee (\mathcal{X} \wedge \mathbb{S})\big) \ \vee\ \mathcal{X} \wedge \mathcal{X}$$

with projections $c \wedge c$, $(c \wedge q,\ q \wedge c)$ and $q \wedge q$. The involution fixes the outer summands and interchanges the two middle ones, so the middle summand is $C_{2+} \wedge \mathcal{X}$. Taking homotopy orbits, with $(C_{2+} \wedge \mathcal{X})_{hC_2} \simeq \mathcal{X}$ and $\mathbb{S}_{hC_2} = \Sigma^\infty BC_{2+}$, gives the splitting of $\Sigma^\infty B_+$. The projection $c \wedge c$ is Σ^∞ of $(M \times M)_+ \to S^0$, so the first summand is projected to by $B \to BC_2$. The splitting of $\Sigma^\infty A_+$ is $\Sigma^\infty BC_{2+} \wedge (\mathbb{S} \vee \mathcal{X})$.

On the level of spaces, $(c \wedge c)\Delta = c$, $(c \wedge q)\Delta = (q \wedge c)\Delta = q$ and $(q \wedge q)\Delta = \bar{\Delta}q$, where $\bar{\Delta}\colon M \to M \wedge M$ is the reduced diagonal. These are equalities of equivariant maps. We precompose with i and $\mathfrak{s}$ and use $ci = 1$, $qi = 0$, $c\mathfrak{s} = 0$, $q\mathfrak{s} = 1$. These relations hold up to point-set homotopies between maps of spectra with trivial action, and such homotopies are equivariant. The components of Δ are therefore as follows: $\mathbb{S} \to \mathbb{S}$ is the identity; $\mathcal{X} \to \mathbb{S}$ is zero; $\mathbb{S} \to C_{2+} \wedge \mathcal{X}$ and $\mathbb{S} \to \mathcal{X} \wedge \mathcal{X}$ are zero; $\mathcal{X} \to \mathcal{X} \wedge \mathcal{X}$ equals $\bar{\Delta}$; and $\mathcal{X} \to C_{2+} \wedge \mathcal{X}$ is the diagonal $\nabla_{\mathcal{X}}$. Passing to homotopy orbits gives the matrix, with $\chi = \bar{\Delta}_{hC_2}$.

The map $x \mapsto (x, x_0)$ is $1 \wedge i\colon \mathcal{Y} \to \mathcal{Y} \wedge \mathcal{Y}$. Its components are $c \wedge ci = c$ into $\mathbb{S}$, $q \wedge ci = q$ into $\mathcal{X} \wedge \mathbb{S}$, and 0 into $\mathbb{S} \wedge \mathcal{X}$ and $\mathcal{X} \wedge \mathcal{X}$. Composed with $\mathfrak{s}$, it is therefore the inclusion of $\mathcal{X}$ as the summand $\mathcal{X} \wedge \mathbb{S}$. The map from $\mathcal{X} \wedge \mathbb{S}$ to $(C_{2+} \wedge \mathcal{X})_{hC_2} \simeq \mathcal{X}$ is the identity. □

Homology through degree three. Let $\mathcal{C} = \mathrm{cofib}(\mathrm{tr}\colon \Sigma^\infty BC_{2+} \to \mathbb{S})$.

Proposition 4.2. (a) *$H_k(\mathcal{C}; \mathbb{Z}) = \mathbb{Z}/2$ for even $k \geq 0$ and 0 for odd k; $H_k(\mathcal{C}; \mathbb{F}_2) = \mathbb{F}_2$ for all $k \geq 0$.*

(b) *There is a map $\Sigma^\infty(B/A) \to \mathcal{C} \wedge \mathcal{X}$ inducing isomorphisms $H_k(B, A; G) \cong H_k(\mathcal{C} \wedge \mathcal{X}; G)$ for $k \leq 3$ and every abelian group G.*

(c) $H_k(B, A; \mathbb{Z}) = H_k(B, A; \mathbb{F}_2) = 0$ *for* $k \le 1$.
(d) *The map* $H_2(M; \mathbb{Z}) \to H_2(B, A; \mathbb{Z})$ *induced by* i_1 *is surjective with kernel* $2H_2(M; \mathbb{Z})$, *and* $H_2(M; \mathbb{F}_2) \to H_2(B, A; \mathbb{F}_2)$ *induced by* i_1 *is an isomorphism.*
(e) $H_3(B, A; \mathbb{Z})$ *is an elementary abelian* 2*-group, so reduction mod* 2 *is injective on it.*

Proof. (a) On H_0, tr_* is the map of coinvariants $\mathbb{Z} \to \mathbb{Z}[C_2]_{C_2} = \mathbb{Z}$ induced by $1 \mapsto 1 + \tau$, that is, multiplication by 2. For $k > 0$ it lands in $H_k(\Sigma^\infty EC_{2+}) = 0$. The group $H_*(BC_2; \mathbb{Z})$ is $\mathbb{Z}$ in degree 0, $\mathbb{Z}/2$ in odd degrees and 0 in positive even degrees. The long exact sequence therefore gives the integral groups. Mod 2, $\mathrm{tr}_* = 0$ and $H_k(\mathcal{C}; \mathbb{F}_2) \cong H_k(\mathbb{S}; \mathbb{F}_2) \oplus H_{k-1}(BC_{2+}; \mathbb{F}_2)$.

(b) By Lemma 4.1, $\Sigma^\infty(B/A) \simeq \mathrm{cofib}(\Sigma^\infty j_+) \simeq \mathrm{cofib}(\Psi)$ with $\Psi = (\mathrm{tr} \wedge 1, \chi) \colon \Sigma^\infty BC_{2+} \wedge \mathcal{X} \to \mathcal{X} \vee D_2\mathcal{X}$. The projection $\mathrm{pr} \colon \mathcal{X} \vee D_2\mathcal{X} \to \mathcal{X}$ satisfies $\mathrm{pr} \circ \Psi = \mathrm{tr} \wedge 1$. It therefore induces $\mathrm{cofib}(\Psi) \to \mathrm{cofib}(\mathrm{tr} \wedge 1) = \mathcal{C} \wedge \mathcal{X}$, with cofibre $\mathrm{cofib}(\mathrm{pr}) = \Sigma D_2\mathcal{X}$. Since M is simply connected, $\mathcal{X}$ is 1-connected, so $\mathcal{X} \wedge \mathcal{X}$ and $D_2\mathcal{X}$ are 3-connected. Hence $H_k(\Sigma D_2\mathcal{X}; G) = 0$ for $k \le 4$.

(c)–(e) Cellular chains of CW spectra are free, so the Künneth sequence for $\mathcal{C} \wedge \mathcal{X}$ is split exact. Since $H_j(\mathcal{X}) = \tilde{H}_j(M) = 0$ for $j \le 1$, part (a) gives $H_k(\mathcal{C} \wedge \mathcal{X}) = 0$ for $k \le 1$,

$$H_2(\mathcal{C} \wedge \mathcal{X}; \mathbb{Z}) = H_0(\mathcal{C}) \otimes H_2(M) = H_2(M) \otimes \mathbb{Z}/2, \qquad H_2(\mathcal{C} \wedge \mathcal{X}; \mathbb{F}_2) = H_2(M; \mathbb{F}_2),$$

$$H_3(\mathcal{C} \wedge \mathcal{X}; \mathbb{Z}) \cong \big(H_3(M) \otimes \mathbb{Z}/2\big) \oplus \mathrm{Tor}\,\big(\mathbb{Z}/2, H_2(M)\big),$$

where we used $H_1(\mathcal{C}; \mathbb{Z}) = 0$. Both summands in degree 3 are killed by 2. By Lemma 4.1, the composite $\mathcal{X} \xrightarrow{(\Sigma^\infty i_{1+})\mathfrak{s}} \Sigma^\infty B_+ \to \Sigma^\infty(B/A) \to \mathcal{C} \wedge \mathcal{X}$ is the inclusion of the bottom cell $\mathbb{S} \to \mathcal{C}$ smashed with $\mathcal{X}$. In degree 2 this composite is the reduction $H_2(M) \to H_2(M) \otimes \mathbb{Z}/2$, respectively the identity of $H_2(M; \mathbb{F}_2)$. □

Cohomology of B through degree three. Let $p \colon EC_2 \times M \times M \to B$ be the double cover, and let $p_!$ be its cohomology transfer.

Lemma 4.3. *For* $1 \le k \le 3$,

$$H^k(B; \mathbb{F}_2) = \mathbb{F}_2\{u^k\} \oplus \{p_!(b \times 1) : b \in H^k(M; \mathbb{F}_2)\},$$

with $i_1^* u^k = 0$, $i_1^* p_!(b \times 1) = b$, $j^* u^k = u^k \otimes 1 \neq 0$ *and* $j^* p_!(b \times 1) = 0$. *Consequently, the image of* $H^3(B, A; \mathbb{F}_2) \to H^3(B; \mathbb{F}_2)$ *is* $\{p_!(b \times 1) : b \in H^3(M; \mathbb{F}_2)\}$.

Proof. By Lemma 4.1 and the 3-connectivity of $D_2\mathcal{X}$, $\dim H^k(B; \mathbb{F}_2) = 1 + \dim H^k(M; \mathbb{F}_2)$ for $1 \le k \le 3$. The map i_1 lifts to the double cover via $x \mapsto (e_0, x, x_0)$, so $i_1^*\lambda$ is trivial and $i_1^* u^k = 0$. The pullback of the cover along i_1 is trivial, with sections $x \mapsto (x, x_0)$ and $x \mapsto (x_0, x)$. Hence $i_1^* p_!(b \times 1) = b + \mathrm{const}_{x_0}^* b = b$, because $k \ge 1$. The pullback of the cover along j is $p' \colon EC_2 \times M \to BC_2 \times M$, and $b \times 1$ pulls back to $p'^*(1 \otimes b)$. By naturality and the projection formula, $j^* p_!(b \times 1) = (1 \otimes b) \cdot p'_!(1) = 0$, because $p'_!(1) = 2 = 0$. Suppose that $n\,u^k + p_!(b \times 1) = 0$ with $n \in \mathbb{F}_2$. Applying i_1^* gives $b = 0$, and then applying j^* gives $n = 0$. Counting dimensions proves the first claim. The last claim follows from exactness of $H^3(B, A; \mathbb{F}_2) \to H^3(B; \mathbb{F}_2) \xrightarrow{j^*} H^3(A; \mathbb{F}_2)$. □

Characteristic classes of z. Let $f \colon M \looparrowright \mathbb{R}^{26}$ be an immersion (Section 2), and let $\nu_+ = \nu_f \oplus \varepsilon$. This is a rank-12 bundle with $TM \oplus \nu_+ \cong \varepsilon^{27}$.

Lemma 4.4. (a) $w_1(z) = 0$.
(b) $j^* z \cong \lambda \otimes \nu_+ - TM$ *stably, and* $w_1(j^* z) = w_2(j^* z) = 0$.
(c) $i_1^* z \cong \varepsilon^{15} - TM$ *stably, so* $w_1(i_1^* z) = 0$ *and* $w_2(i_1^* z) = v$.

Proof. Over A, the involution splits $T_xM \oplus T_xM$ into its (±1)-eigenspaces, so $j^*T = TM \oplus (TM \otimes \lambda)$. For a bundle U' of rank m,

$$w_1(U' \otimes \lambda) = w_1(U') + mu, \qquad w_2(U' \otimes \lambda) = w_2(U') + (m-1)w_1(U')u + \tbinom{m}{2}u^2. \tag{4.5}$$

(a) Since $H^1(M;\mathbb{F}_2) = 0$, Lemma 4.3 gives $H^1(B;\mathbb{F}_2) = \mathbb{F}_2\{u\}$, and j^* is injective on this group. As $w_1(j^*T) = w_1(M) + w_1(M) + 15u = u$, we obtain $w_1(T) = u$ and hence $w_1(z) = 27u + u = 0$.

(b) We have $27\lambda = (TM \oplus \nu_+) \otimes \lambda$, so $j^*z = 27\lambda - TM - TM \otimes \lambda + 3 = \nu_+ \otimes \lambda - TM + 3$. Applying (4.5) with $m = 12$ and $w_1(\nu_+) = 0$ gives $w_1(\nu_+ \otimes \lambda) = 12u = 0$ and $w_2(\nu_+ \otimes \lambda) = w_2(\nu_+) + 66u^2 = w_2(\nu_+)$. Since $w(-TM) = w(\nu_+)$,

$$w_1(j^*z) = 0, \qquad w_2(j^*z) = w_2(\nu_+ \otimes \lambda) + w_1(\nu_+ \otimes \lambda)w_1(\nu_+) + w_2(\nu_+) = 2w_2(\nu_+) = 0.$$

(c) The map i_1 lifts to the double cover, so $i_1^*\lambda$ is trivial and $i_1^*T = TM \oplus T_{x_0}M \cong TM \oplus \varepsilon^{15}$. Thus $i_1^*z = \varepsilon^{27} - TM - \varepsilon^{15} + \varepsilon^3$ and $w(i_1^*z) = w(M)^{-1}$. The degree-2 part of $w(M)^{-1}$ is $w_1(M)^2 + w_2(M) = v$. □

Corollary 4.6. *R is 1-connected and has a CW structure with no cells of dimension < 2. Let*

$$\theta\colon H_2(M;\mathbb{F}_2) = H_2(M;\mathbb{Z}) \otimes \mathbb{F}_2 \xrightarrow{\cong} H_2(B,A;\mathbb{Z}) \cong H_2(R;\mathbb{Z}) \xleftarrow[\cong]{h} \pi_2(R),$$

where the first isomorphism is induced by i_1 (Proposition 4.2(d)), the second is the Thom isomorphism and h is the Hurewicz map. In particular, $\pi_2(R)$ is an $\mathbb{F}_2$-vector space. For $\omega \in \pi_2(R)$ we write $\langle v, \omega\rangle = \langle v, \theta^{-1}(\omega)\rangle \in \mathbb{F}_2$.

Proof. By Lemma 4.4(a), the Thom isomorphism holds with $\mathbb{Z}$ coefficients. Hence $H_k(R;\mathbb{Z}) \cong H_k(B,A;\mathbb{Z})$, which vanishes for $k \le 1$ by Proposition 4.2(c). The spectrum R is bounded below, so it is 1-connected and h is an isomorphism in degree 2. By Lemma 3.5, $R \simeq (W,\partial W)^{z_W}$ with $z_W = 27\lambda - TW + 3$. Since W is compact, $z_W = \zeta - \varepsilon^m$ for a vector bundle ζ of rank $m \ge 2$. Then $(W,\partial W)^{z_W} = \Sigma^{-m}\Sigma^\infty\big(\operatorname{Th}(\zeta)/\operatorname{Th}(\zeta|_{\partial W})\big)$. Here $\operatorname{Th}(\zeta)/\operatorname{Th}(\zeta|_{\partial W})$ is a finite complex with one 0-cell and all other cells of dimension $\ge m \ge 2$, so it is simply connected. Fact 2.9 gives the cell structure. □

5. The Hurewicz image of the obstruction

Lemma 5.1. *Let $f\colon M \looparrowright \mathbb{R}^{26}$ be an immersion. After a C^∞-small perturbation, f is self-transverse, and*

$$S_f = \{(x,y) \in M \times M : x \ne y,\ f(x) = f(y)\}$$

is a closed 4-manifold with free involution τ. Put $Q = S_f/C_2$, let $p_Q\colon S_f \to Q$ be the double cover, and let $\lambda_Q = S_f \times_{C_2} \mathbb{R}^-$. There is $\ell \in C^\infty(M,\mathbb{R})$ such that 0 is a regular value of

$$\phi\colon S_f \to \mathbb{R}, \qquad \phi(x,y) = \ell(x) - \ell(y).$$

For such ℓ, $g = (f,\ell)\colon M \looparrowright \mathbb{R}^{27}$ is a self-transverse immersion with $\widetilde{D}_g = \phi^{-1}(0)$. Moreover, $D_g \subset Q$ is the zero set of the section $\bar\phi$ of λ_Q induced by the odd function ϕ, and this section is transverse to zero.

Proof. By the multijet transversality theorem [GG73, Ch. II, Thm. 4.13], applied to the diagonal of $\mathbb{R}^{26} \times \mathbb{R}^{26}$, a C^∞-small perturbation of f is a self-transverse immersion. Then S_f is a submanifold of dimension $2 \cdot 15 - 26 = 4$. It is compact because f is locally injective, and τ acts freely on it.

For each $(x,y) \in S_f$, choose $\ell_{(x,y)} \in C^\infty(M,\mathbb{R})$ with $\ell_{(x,y)}(x) \neq \ell_{(x,y)}(y)$. This inequality persists on a neighbourhood of (x,y), so by compactness finitely many $\ell_1,\dots,\ell_k$ suffice. The map

$$\Phi\colon S_f \times \mathbb{R}^k \to \mathbb{R}, \qquad \Phi\big((x,y),t\big) = \sum_i t_i\big(\ell_i(x) - \ell_i(y)\big)$$

is a submersion. By the parametric transversality theorem [GP74, Ch. 2, §3], 0 is therefore a regular value of $\Phi(\cdot,t)$ for almost every t. We take $\ell = \sum_i t_i\ell_i$ for such a t.

Now $g(x) = g(y)$ if and only if $(x,y) \in S_f$ and $\phi(x,y) = 0$. At such a point, $ds_g = (ds_f, d\phi)\colon T_xM \oplus T_yM \to \mathbb{R}^{26} \times \mathbb{R}$. Since ds_f is surjective with kernel $T_{(x,y)}S_f$, the map ds_g is surjective if and only if $d\phi$ is nonzero on $T_{(x,y)}S_f$. This holds because 0 is a regular value of ϕ, so g is self-transverse. As $\phi \circ \tau = -\phi$, ϕ defines a section $\bar\phi$ of λ_Q, transverse to zero, with zero set $\phi^{-1}(0)/C_2 = D_g$. □

Proposition 5.2. *For $g = (f,\ell)$ as in Lemma 5.1, the Hurewicz image of $o(g)$ in $H_3(R;\mathbb{Z})$ is zero.*

Proof. Let κ_D be as in Definition 3.8. By Fact 2.3, under the Thom isomorphism the Hurewicz image of $\mu(g)$ is $(\kappa_D)_*[D_g] \in H_3(B;\mathbb{Z})$. The Hurewicz image of $o(g)$ is therefore the image of $(\kappa_D)_*[D_g]$ in $H_3(B,A;\mathbb{Z})$. By Proposition 4.2(e), it suffices to show that the image of $(\kappa_D)_*[D_g]_2$ in $H_3(B,A;\mathbb{F}_2)$ vanishes. Equivalently, $\langle \beta, (\kappa_D)_*[D_g]_2\rangle = 0$ for every $\beta \in H^3(B,A;\mathbb{F}_2)$, evaluated through its image in $H^3(B;\mathbb{F}_2)$. By Lemma 4.3, that image is $p_!(b \times 1)$ for some $b \in H^3(M;\mathbb{F}_2)$.

An equivariant map $S_f \to EC_2$ extending the chosen map on $\widetilde{D}_g$ defines $\kappa\colon Q \to B$ with $\kappa|_{D_g} = \kappa_D$. Then p_Q is the pullback of p, and $\kappa^*\lambda = \lambda_Q$. Since D_g is the transverse zero set of a section of λ_Q, its mod 2 fundamental class is Poincaré dual in Q to $w_1(\lambda_Q) = \kappa^*u$. By naturality of the transfer and the projection formula,

$$\langle p_!(b\times 1), (\kappa_D)_*[D_g]_2\rangle = \langle \kappa^* p_!(b\times 1) \cup \kappa^* u, [Q]_2\rangle = \langle (p_Q)_!\big(b' \cup p_Q^*\kappa^* u\big), [Q]_2\rangle,$$

where b' is the pullback of $b \times 1$ to S_f. Since $p_Q^*\lambda_Q$ is trivial, $p_Q^*\kappa^* u = 0$, and the pairing vanishes. □

6. Vanishing of the obstruction

Models over S^2. Let $\sigma\colon (S^2,*) \to (M,x_0)$ be a based map. We apply the constructions of Sections 3 and 4 to the closed simply connected manifold S^2 and put

$$B_\sigma = EC_2 \times_{C_2} (S^2 \times S^2), \qquad A_\sigma = BC_2 \times S^2.$$

Let $\hat\sigma\colon (B_\sigma, A_\sigma) \to (B,A)$ be induced by $\sigma\times\sigma$. Put $z_\sigma = \hat\sigma^* z$, $\mathcal{E}_\sigma = B_\sigma^{z_\sigma}$, $A_\sigma^z = A_\sigma^{j_\sigma^* z_\sigma}$ and $R_\sigma = (B_\sigma, A_\sigma)^{z_\sigma}$, with boundary map δ_σ. Then $\hat\sigma$ induces a map from the cofibre sequence $A_\sigma^z \to \mathcal{E}_\sigma \to R_\sigma \xrightarrow{\delta_\sigma} \Sigma A_\sigma^z$ to $A^z \to \mathcal{E} \to R \xrightarrow{\delta} \Sigma A^z$. This map sends bottom classes ι_0 to bottom classes. Let

$$n_\sigma = \langle v, \sigma_*[S^2]\rangle = \langle w_2(\sigma^*TM), [S^2]\rangle \in \mathbb{F}_2.$$

Lemma 6.1. (a) *$H_*(R_\sigma;\mathbb{Z})$ is $\mathbb{Z}/2$, 0, $\mathbb{Z}$ in degrees 2, 3, 4 and zero in all other degrees. Hence $\pi_2(R_\sigma) = \mathbb{Z}/2\{\omega_\sigma\}$, and $\hat\sigma_*(\omega_\sigma) = \theta\big(\rho_2\sigma_*[S^2]\big)$, where ρ_2 is reduction mod 2.*

(b) *$R_\sigma \simeq \Sigma^2\mathbb{S}/2 \cup_{\psi_\sigma} e^4$ for some $\psi_\sigma \in \pi_3(\Sigma^2\mathbb{S}/2)$, with bottom cell ω_σ.*

(c) *$H^2(R_\sigma;\mathbb{F}_2) = \mathbb{F}_2\{U\cdot a\}$ for some $a \in H^2(B_\sigma, A_\sigma;\mathbb{F}_2)$ with $a^2 \neq 0$ in $H^4(B_\sigma, A_\sigma;\mathbb{F}_2)$.*

(d) *If $n_\sigma = 0$, then $z_\sigma \cong 12\lambda - \varepsilon^{12}$ stably; in particular, $w_1(z_\sigma) = w_2(z_\sigma) = 0$.*

Proof. (a) By Lemma 3.5 for S^2, (B_σ, A_σ) has the relative homology of $(\mathrm{SP}^2\, S^2, \Delta S^2)$. Identifying an unordered pair of points of $\mathbb{CP}^1$ with the binary quadratic form that vanishes on it gives $\mathrm{SP}^2(\mathbb{CP}^1) \cong \mathbb{CP}^2$. Under this identification the diagonal becomes the conic Γ of forms with a double root. As Γ has degree 2, $H_2(\Gamma;\mathbb{Z}) \to H_2(\mathbb{CP}^2;\mathbb{Z})$ is multiplication by 2. The exact sequence of $(\mathbb{CP}^2, \Gamma)$ then gives the stated groups. Since $w_1(z_\sigma) = \hat\sigma^* w_1(z) = 0$, the Thom isomorphism transfers these groups to R_σ. Hence R_σ is 1-connected with $\pi_2(R_\sigma) \cong H_2(R_\sigma;\mathbb{Z}) = \mathbb{Z}/2$. By Proposition 4.2(d) for S^2, $H_2(B_\sigma, A_\sigma;\mathbb{Z})$ is generated by the image of $[S^2]$ under i_1^σ. Because σ is based, $\hat\sigma \circ i_1^\sigma = i_1 \circ \sigma$. Naturality of the Hurewicz and Thom isomorphisms then gives $\hat\sigma_*(\omega_\sigma) = \theta(\rho_2 \sigma_*[S^2])$.

(b) As in Corollary 4.6, R_σ is a desuspension of a simply connected finite complex. Fact 2.9 gives a CW structure with one cell in each of the dimensions $2, 3, 4$, in which the 3-cell is attached with degree 2.

(c) The restriction $H^2(\mathbb{CP}^2;\mathbb{F}_2) \to H^2(\Gamma;\mathbb{F}_2)$ is zero because Γ has even degree, and $H^1(\Gamma) = H^3(\Gamma) = H^4(\Gamma) = 0$. Hence $H^k(\mathbb{CP}^2, \Gamma;\mathbb{F}_2) \to H^k(\mathbb{CP}^2;\mathbb{F}_2)$ is an isomorphism for $k = 2, 4$, compatible with cup products. The square of the generator of $H^2(\mathbb{CP}^2;\mathbb{F}_2)$ is nonzero, and Lemma 3.5 transports this fact to (B_σ, A_σ).

(d) The bundle $\sigma^* TM$ is oriented of rank 15 over S^2. Such bundles are classified by $\pi_1(SO(15)) = \mathbb{Z}/2$, which is detected by w_2. If $n_\sigma = 0$, the bundle is trivial, so $\hat\sigma^* T$ is the Borel construction on $\varepsilon^{15} \times \varepsilon^{15}$, which is $\varepsilon^{15} \oplus 15\lambda$. Then $z_\sigma = 27\lambda - \varepsilon^{15} - 15\lambda + \varepsilon^3 = 12\lambda - \varepsilon^{12}$ and $w(z_\sigma) = (1+u)^{12} = (1+u^4)(1+u^8)$. □

Elements of $\eta \cdot \pi_2(R)$. Every $\omega \in \pi_2(R)$ has the form $\omega = \hat\sigma_*(\omega_\sigma)$ with $n_\sigma = \langle v, \omega\rangle$. To see this, write $\theta^{-1}(\omega) = \rho_2 \alpha$ with $\alpha \in H_2(M;\mathbb{Z})$. Since M is simply connected, the Hurewicz theorem represents $\alpha = \sigma_*[S^2]$ by a based map, and Lemma 6.1(a) applies.

Proposition 6.2. *If $\omega \in \pi_2(R)$ and $\langle v, \omega\rangle = 0$, then $\eta\omega = 0$ in $\pi_3(R)$.*

Proof. Write $\omega = \hat\sigma_*(\omega_\sigma)$ with $n_\sigma = 0$. It suffices to show that $\eta\omega_\sigma = 0$ in $\pi_3(R_\sigma)$. By Lemma 6.1(c),(d) and (2.4),

$$\mathrm{Sq}^2(U \cdot a) = U \cdot \mathrm{Sq}^2\, a = U \cdot a^2 \neq 0.$$

Consider the cell structure of Lemma 6.1(b). If $\psi_\sigma = 0$, then $R_\sigma \simeq \Sigma^2 \mathbb{S}/2 \vee S^4$ and the class $U \cdot a$ is pulled back along the retraction onto $\Sigma^2\mathbb{S}/2$. Since $H^4(\Sigma^2\mathbb{S}/2;\mathbb{F}_2) = 0$, this would force $\mathrm{Sq}^2(U \cdot a) = 0$. Hence $\psi_\sigma \neq 0$, and Lemma 2.7 gives $\psi_\sigma = \eta\, \omega_\sigma$. An attaching map is null in the resulting complex, so $\eta\omega_\sigma = 0$. □

Proposition 6.3. *Let $\iota_0 \in \pi_0(A^z)$ be the bottom class. Then $\eta\, \iota_0 \neq 0$, $\eta^2 \iota_0 \neq 0$, and*

$$\delta\omega = \langle v, \omega\rangle\, \eta\, \iota_0 \qquad \text{for every } \omega \in \pi_2(R).$$

Proof. By Lemma 4.4(b), $j^* z$ is spin, so Lemma 2.6 gives $\eta\, \iota_0 \neq 0$ and $\eta^2 \iota_0 \neq 0$. The same holds for the bottom class of A^z_σ, because $j_\sigma^* z_\sigma$ is pulled back from $j^* z$.

Write $\omega = \hat\sigma_*(\omega_\sigma)$ with $n_\sigma = \langle v, \omega\rangle$. Then $\delta\omega = \hat\sigma_*(\delta_\sigma \omega_\sigma)$ and $\hat\sigma_*(\eta\, \iota_0) = \eta\, \iota_0$. It therefore suffices to show that $\delta_\sigma \omega_\sigma = n_\sigma\, \eta\, \iota_0$ in $\pi_1(A^z_\sigma)$.

The composite $A_\sigma \to B_\sigma \to BC_2$ is the projection $BC_2 \times S^2 \to BC_2$, and $B_\sigma \to BC_2$ has simply connected fibre $S^2 \times S^2$. Hence $H_1(A_\sigma;\mathbb{Z}) \to H_1(B_\sigma;\mathbb{Z})$ is an isomorphism, and the connecting map $H_2(B_\sigma, A_\sigma;\mathbb{Z}) \to H_1(A_\sigma;\mathbb{Z})$ is zero. By the Thom isomorphism, $\delta_\sigma\omega_\sigma$ has zero Hurewicz image in $H_1(A^z_\sigma;\mathbb{Z})$. With the product CW structure on $BC_2 \times S^2$, A^z_σ has cells only in dimensions ≥ 0 and a single 0-cell. Thus $\pi_0(A^z_\sigma) = \mathbb{Z}\{\iota_0\}$, and Lemma 2.8 gives $\delta_\sigma\omega_\sigma \in \{0, \eta\, \iota_0\}$. Let $\iota_0^{\mathcal{E}_\sigma} = (j_\sigma)_* \iota_0$ be the bottom class of $\mathcal{E}_\sigma$. Since $\pi_2(R_\sigma) = \{0, \omega_\sigma\}$ and $\eta\, \iota_0 \neq 0$, exactness of the sequence (3.6) for σ shows that $\delta_\sigma\omega_\sigma = \eta\, \iota_0$ if and only if $\eta\, \iota_0^{\mathcal{E}_\sigma} = 0$ in $\pi_1(\mathcal{E}_\sigma)$.

If $n_\sigma = 0$, then z_σ is spin by Lemma 6.1(d). Lemma 2.6 then gives $\eta\, \iota_0^{\mathcal{E}_\sigma} \neq 0$, and hence $\delta_\sigma \omega_\sigma = 0$.

If $n_\sigma = 1$, consider $i_1^\sigma \colon S^2 \to B_\sigma$. Since $\hat{\sigma} \circ i_1^\sigma = i_1 \circ \sigma$, Lemma 4.4(c) gives $w_2\big((i_1^\sigma)^* z_\sigma\big) = \sigma^* v \neq 0$. By Lemma 2.5(b), $\eta\, \iota_0 = 0$ in $\pi_1\big((S^2)^{(i_1^\sigma)^* z_\sigma}\big)$. The induced map of Thom spectra carries this bottom class to $\iota_0^{\mathcal{E}_\sigma}$. Hence $\eta\, \iota_0^{\mathcal{E}_\sigma} = 0$ and $\delta_\sigma \omega_\sigma = \eta\, \iota_0$. □

Corollary 6.4. *If $y \in \eta \cdot \pi_2(R)$ and $\delta y = 0$, then $y = 0$.*

Proof. Write $y = \eta\omega$. Since δ commutes with η, Proposition 6.3 gives $0 = \delta y = \eta\, \delta\omega = \langle v, \omega \rangle\, \eta^2 \iota_0$. As $\eta^2 \iota_0 \neq 0$, we obtain $\langle v, \omega \rangle = 0$. Proposition 6.2 then gives $y = \eta\omega = 0$. □

Remark: We now prove Theorem 2.1. By Cohen's theorem [Coh85], there is an immersion $f \colon M \looparrowright \mathbb{R}^{26}$. Choose f and ℓ as in Lemma 5.1, so that $g = (f, \ell) \colon M \looparrowright \mathbb{R}^{27}$ is a self-transverse immersion. Let $o = o(g) = \varrho_* \mu(g) \in \pi_3(R)$.

(1) By Proposition 5.2, the Hurewicz image of o in $H_3(R; \mathbb{Z})$ is zero.
(2) By Corollary 4.6, R has a CW structure with no cells of dimension < 2. Lemma 2.8 with $k = 2$ therefore gives $o \in \eta \cdot \pi_2(R)$.
(3) By exactness of (3.6), $\delta o = \delta \varrho_* \mu(g) = 0$.
(4) By Corollary 6.4, $o = 0$.
(5) By Corollary 3.10, M embeds smoothly in $\mathbb{R}^{27}$. □

Consider the closed manifold

$$N = S^1 \times \mathbb{RP}^2 \times \mathbb{RP}^4 \times \mathbb{RP}^8.$$

It is a closed smooth 15-manifold, and its normal Stiefel–Whitney class of degree 11 is

$$\bar{w}_{11}(N) = b\, c^3 d^7 \neq 0,$$

where b, c and d are the degree-1 generators of the three real projective space factors. If a closed 15-manifold embeds in $\mathbb{R}^{26}$, its normal Stiefel–Whitney class of degree 11 must vanish. Hence N does not embed in $\mathbb{R}^{26}$, and therefore it does not embed in $\mathbb{R}^{25}$ either. For related non-embedding results, see the survey [Sko07]. The manifold N is not simply connected. It therefore does not bear on whether Theorem 2.1 is optimal; it only shows that, for general closed 15-manifolds, the dimension 27 cannot be lowered. Thus 27 is a lower bound for a Euclidean dimension in which all closed 15-manifolds embed, and what remains open is the upper bound: whether every closed 15-manifold that is not simply connected embeds in $\mathbb{R}^{27}$. In my view this is the case (I am attempting to prove it). If so, 27 would be the optimal dimension for all closed 15-manifolds, which would settle the case $n = 15$ of the embedding conjecture that every closed n-manifold embeds in $\mathbb{R}^{2n+1-\alpha(n)}$ [Sko07].

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